HACKDAVIS 2026 · UC DAVIS

CARDIAC MONITORING / V0.1

CardioSense

A wearable ECG patch that catches the arrhythmia your doctor missed — because it wasn't there when you walked in.

CARDIOSENSE · HACKDAVIS 2026

1 in 4

STROKES · WORLDWIDE

strokes worldwide are caused by atrial fibrillation —

a heart rhythm that millions of people have, often without ever knowing it. AFib is silent in the majority of people who carry it, and it is the single most preventable cause of stroke we still routinely miss.

Atrial fibrillation comes in three forms.

Two are easy to catch. One is not.

01 ● caught easily

Permanent AFib

The heart is in arrhythmia all the time. Any standard 12-lead ECG, anywhere, will catch it on the first read. Treatment is well-established.

02 ● caught easily

Persistent AFib

Episodes last more than seven days and don't self-resolve. The patient is still in arrhythmia when they walk into the clinic. The ECG catches it.

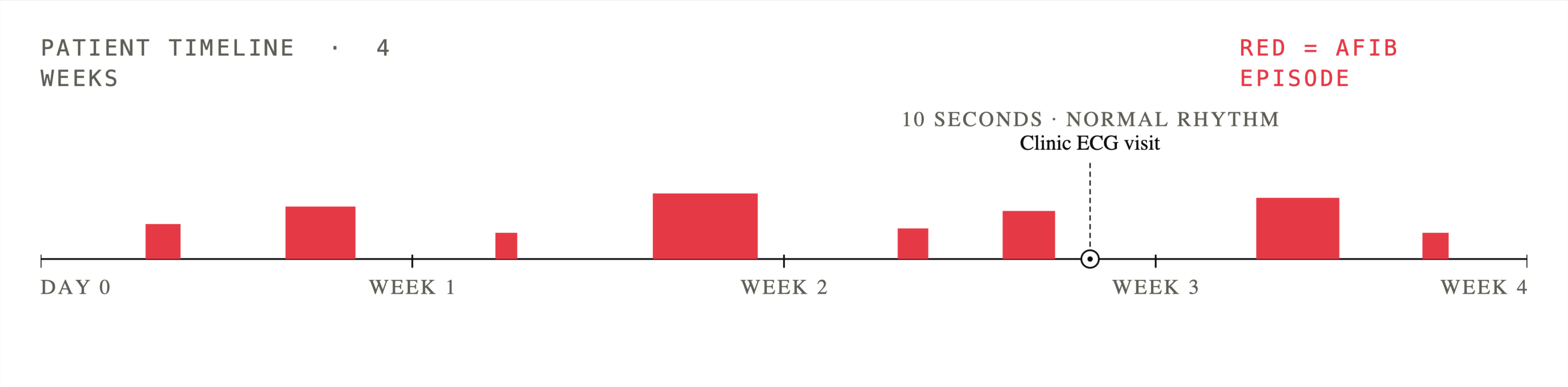
03 OUR TARGET

Paroxysmal AFib

Episodes come and go on their own — minutes, hours, occasionally days. By the time the patient sees a cardiologist, the rhythm is normal again. A 10-second clinic ECG misses it.

Paroxysmal AFib hides between appointments.

A 10-second ECG cannot catch what doesn't show up for it.



By the time the patient arrives, the rhythm is back to normal. They go home undiagnosed. Weeks later, an episode causes a stroke. The only way to catch paroxysmal AFib is to watch continuously.

The tools that exist were not built for who needs them most.

EXISTING SOLUTION	COST	ACCESS	TIME TO RESULT
12-lead ECG, in clinic	covered, with insurance	misses paroxysmal AFib by design	10 seconds, in person
Holter / patch monitor	$300–$1,500	physician's order required	days to weeks to read out
Apple Watch / Fitbit	$400+ device	on-demand only, English-only UI	seconds, episodic
CardioSense	**< $30 in materials**	**no prescription, no cloud, no insurance**	**<10 seconds, continuously**

If you are uninsured, none of the existing options exist for you.

49

years

CALIFORNIA · CENTRAL VALLEY

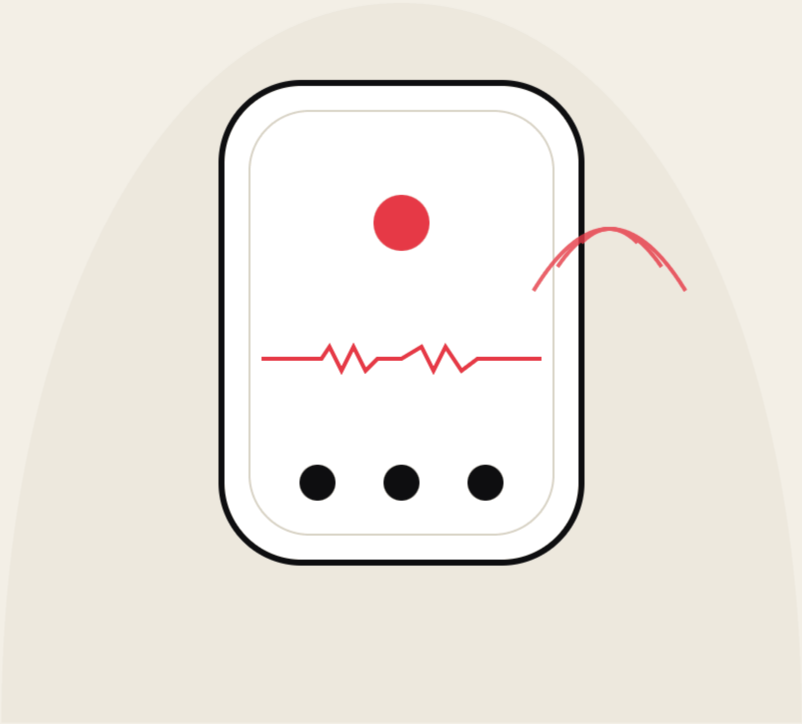
is the average life expectancy of a California farmworker.

That is 28 years below the state average. Heart disease is a leading contributor. There are an estimated **500,000 to 800,000 farmworkers** in California — most uninsured, most Spanish-speaking, most working **90+ minutes from the nearest cardiologist**.

> A $400 consumer wearable is not the answer for this population. That is who we built CardioSense for.

CardioSense

A continuous cardiac monitor that runs end-to-end on a laptop and a phone. The verdict on your rhythm is on your screen in under ten seconds.



— 01

No cloud

Inference runs locally. Nothing leaves the device.

— 02

Minimal subscription

Hardware sold at cost. No data liability.

— 03

No prescription

Patient deploys it. Clinic receives the stream.

Four layers, end to end.

Each stage emits a contract the next stage consumes.

— 01

Hardware

Three electrodes capture cardiac electrical activity. An analog front end amplifies the signal. An Arduino samples at 250 Hz and transmits over BLE.

→ 24-byte BLE NOTIFY / 40 ms

→

— 02

Signal processing

Bandpass + notch filtering, baseline-wander correction, Pan-Tompkins R-peak detection, HRV feature extraction over 10-second windows.

→ stage2/shared/input.json

→

— 03

Machine learning

A 1D convolutional neural network classifies each 10-second window as Normal or Arrhythmia, with explicit Uncertain gating.

→ shared/prediction.json

→

— 04

Two interfaces

A SwiftUI app for the patient. A Streamlit dashboard for the clinician. Both poll the same JSON every two seconds.

→ patient & clinic

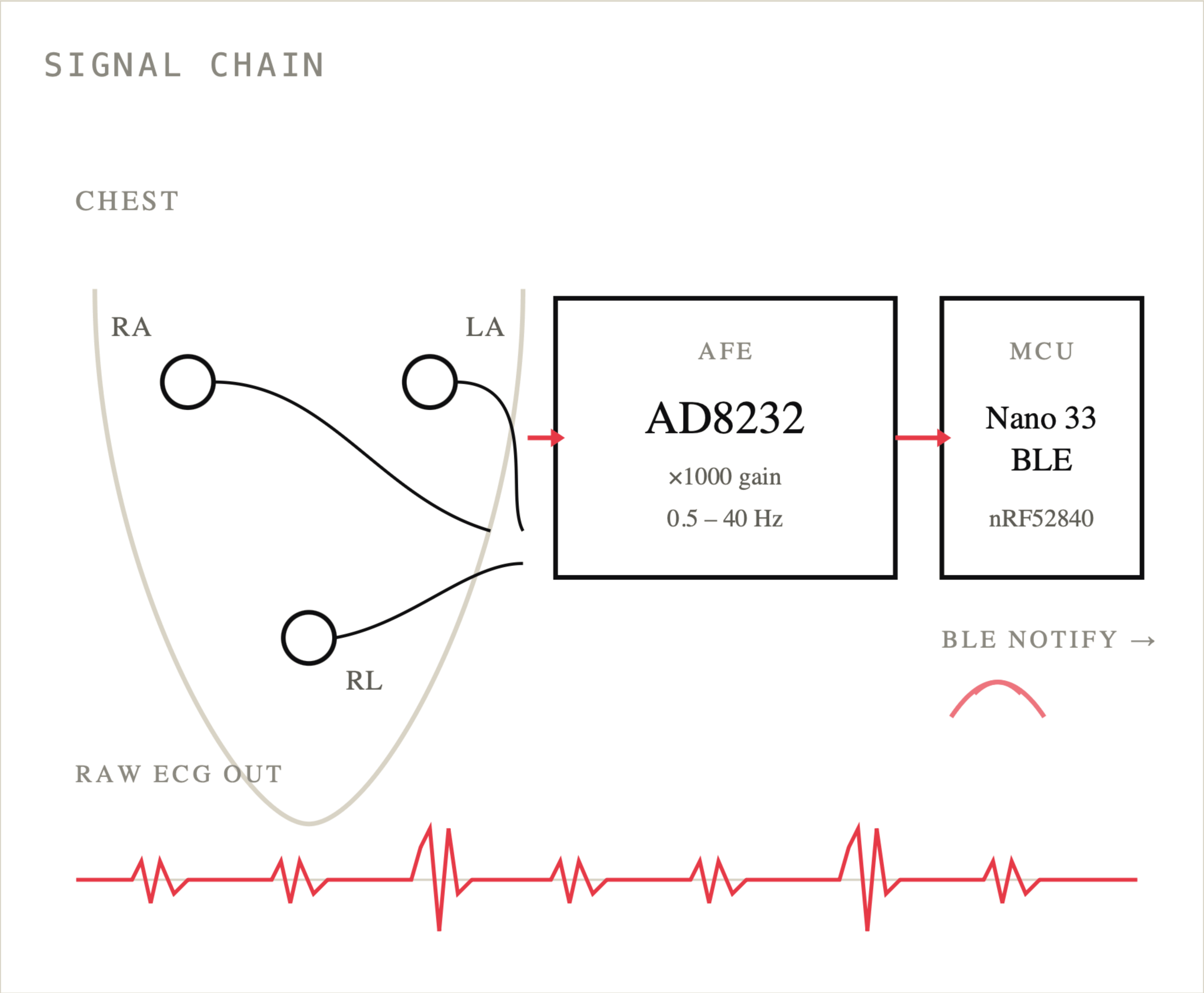
No cloud. No SDK. Every layer runs locally on commodity hardware. The dataflow above is the real one — same JSON contracts you'd find if you cloned the repo.

— UNDER THE HOOD

How it actually works.

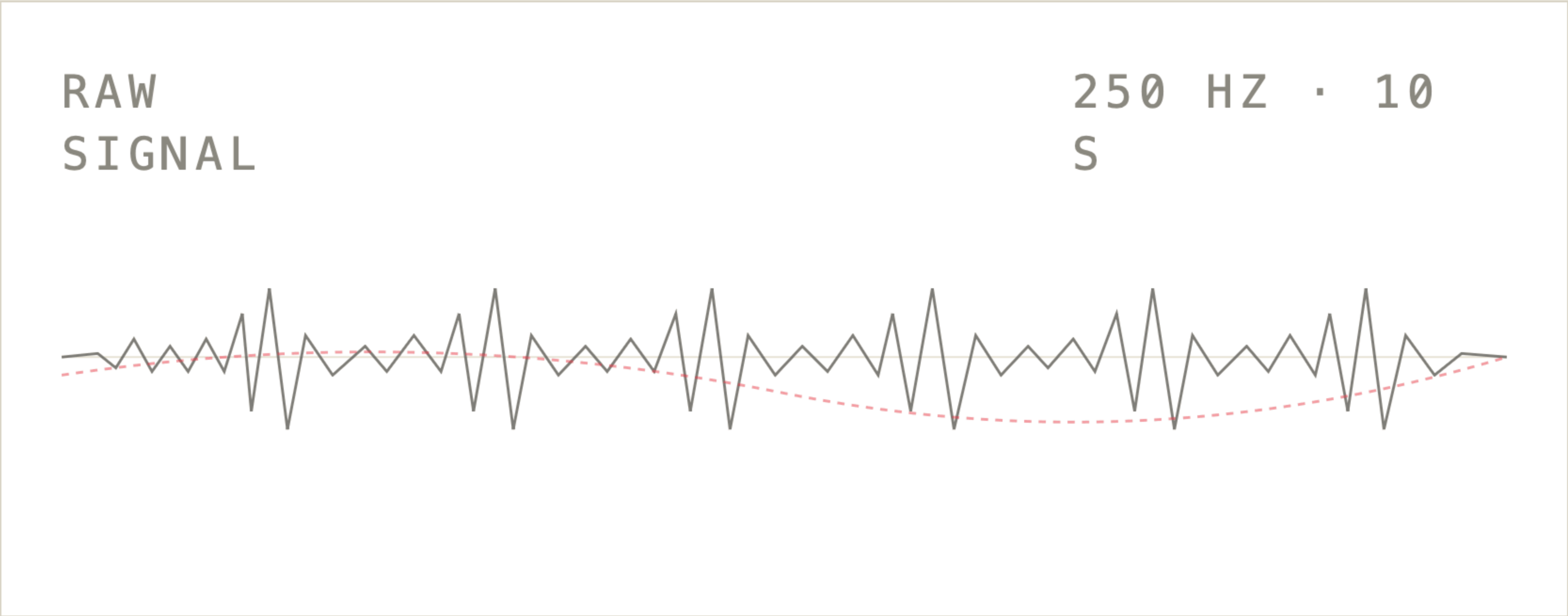
Four layers, in order. From the signal coming off three gel electrodes on a chest, to the verdict on a phone screen — in under ten seconds.

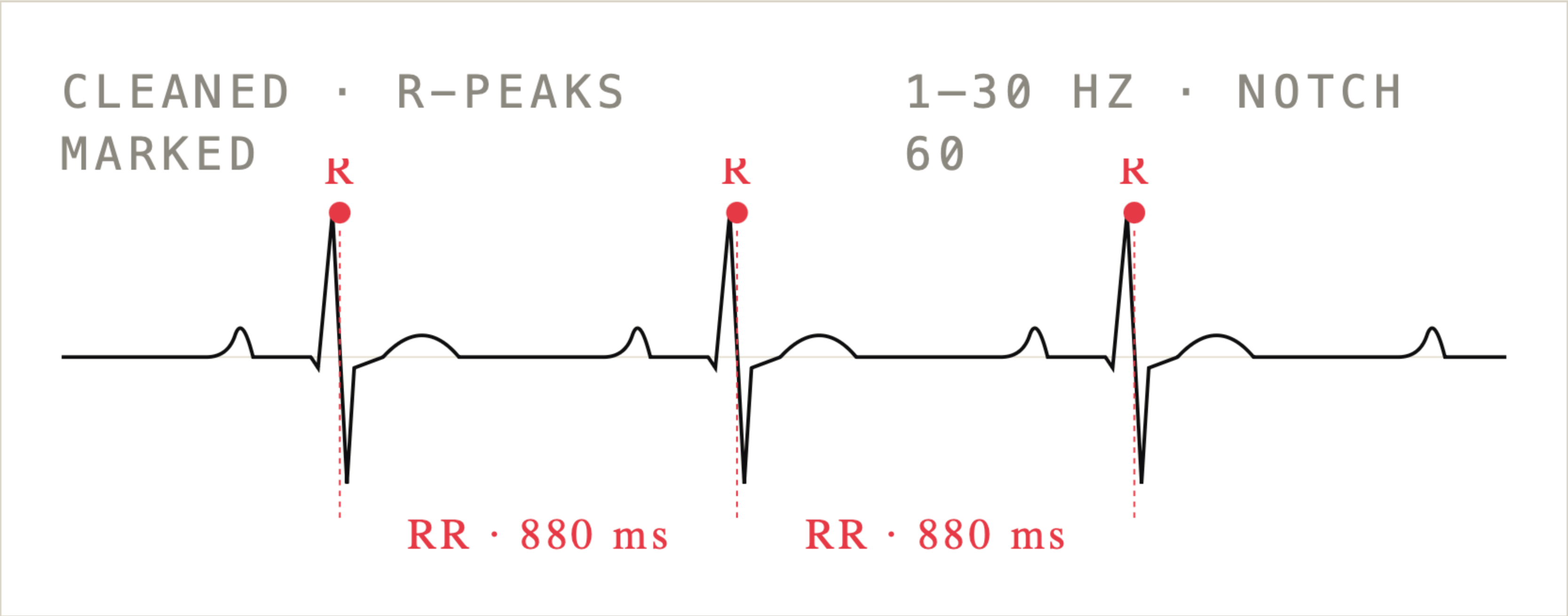
Three electrodes, one front end, one radio.



MCU	**Arduino Nano 33 BLE** nRF52840, 250 Hz ADC, BLE NOTIFY radio
AFE	**SparkFun AD8232** single-lead breakout, ×1000 gain, 0.5–40 Hz bandpass
ELECTRODES	**3-lead wet gel** RA, LA, RL driven ground
TRANSPORT	**24-byte BLE payload** 10 samples + metadata per 40 ms packet
BOM	**under $30** measured retail, not at-scale procurement

Clean the signal. Find the beat. Measure variability.





— 01

Bandpass
4th-order Butterworth · 1–30 Hz

— 02

Notch + baseline
60 Hz notch · 600 ms wander correction

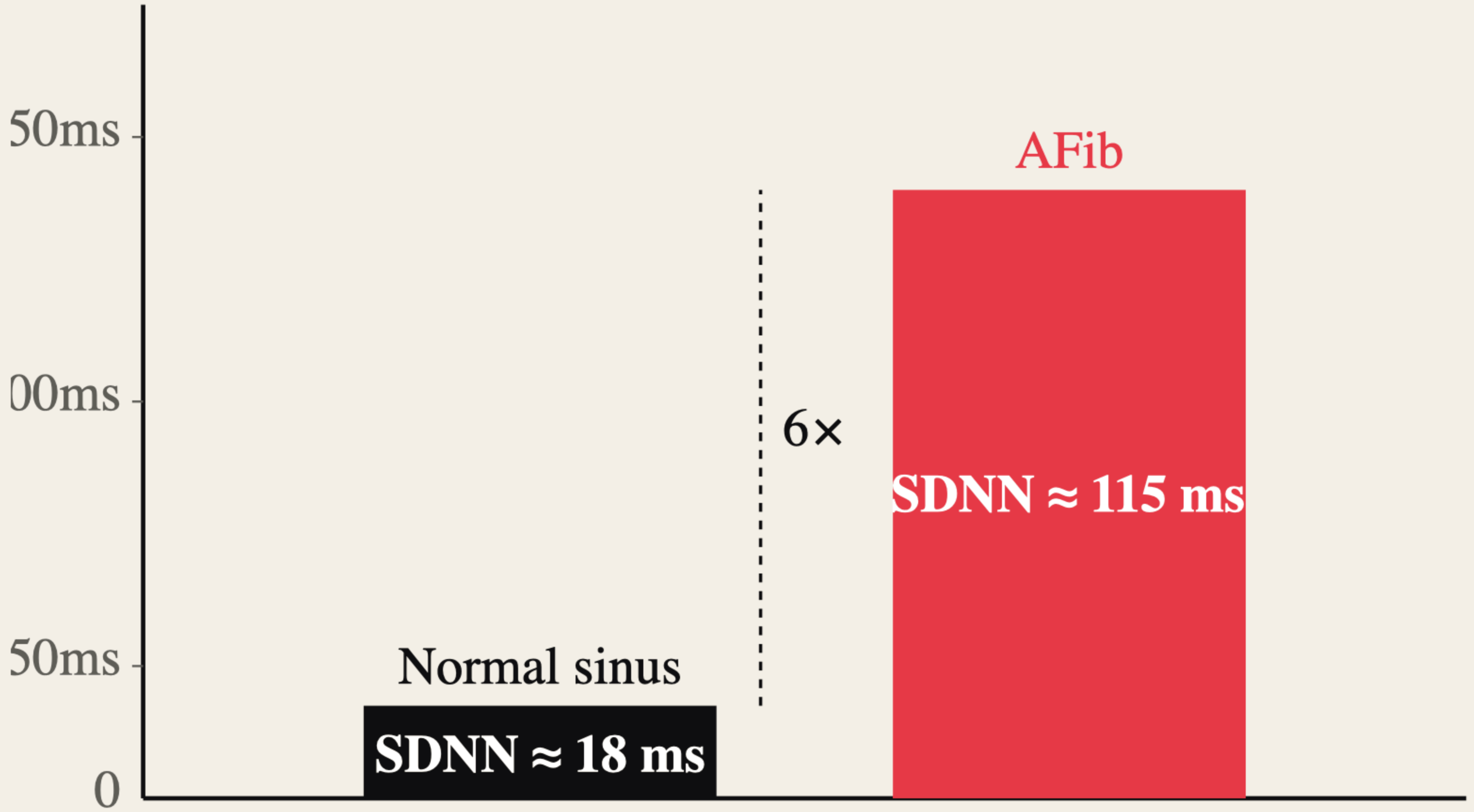
— 03

Pan-Tompkins
diff · square · 150 ms integrate · adaptive threshold

— 04

HRV features
SDNN · RMSSD · pNN50 · per 10 s window

The classifier rides on a 6× gap in heart-rate variability.



In a healthy heart, beat-to-beat intervals are regular. In AFib, the atria are firing chaotically and the intervals lose their structure.

SDNN	standard deviation of NN intervals overall variability
RMSSD	root mean square of successive differences short-term variability — primary AFib signal
PNN50	% of NN pairs differing > 50 ms beat irregularity index

> That ratio is what the CNN classifies on. No ratio, no model.

A 43k-parameter 1D CNN.

10 ms per inference. 200 KB on disk.

ARCHITECTURE · (B, 1, 2500) → (B, 2)

ECG 1×2500 → Conv1D ×4 +BN +ReLU +pool +dropout → GAP → Dense → Softmax → [Normal, Arrhythmia]

PARAMETERS
43,362
total trainable

INPUT WINDOW
10 s
250 Hz, single channel, 3 s stride

INFERENCE
~10 ms
Apple MPS · 25–35 ms on CPU

MODEL SIZE
200 KB
small enough to ship on a watch

Trained on 71 clinical patient records — MIT-BIH Atrial Fibrillation Database + MIT-BIH Arrhythmia Database, resampled to 250 Hz. Splits are at the record level, not the window level, so no patient appears in both train and test.

97%

accuracy on 100 stratified held-out clips. 95% with the strict Uncertain gate engaged as a safety filter.

HELD-OUT RECORDS · MIT-BIH

CLASS	PRECISION	RECALL	F1	SUPPORT
Normal	0.913	0.926	0.919	1,536
Arrhythmia	0.904	0.887	0.895	1,205
Accuracy	**–**	**–**	**0.909**	**2,741**

Both class recalls clear the ≥0.85 hackathon target. Validation script ships in the repo as `scripts.validate_100`.

When the model doesn't know, it says so.

Most consumer wearables collapse low-confidence predictions into the closest binary label. CardioSense doesn't. Three runtime gates trip an explicit Uncertain state instead of forcing a verdict.

→ 01 **Model confidence < 60%**
Softmax probability of the winning class below threshold.

→ 02 **Fewer than 3 R-peaks detected**
HRV is undefined. We do not extrapolate.

→ 03 **Signal flat or low quality**
Electrode contact lost, motion artifact, baseline collapse.

FROM THE IOS REFERENCE TAB

"Intended for study and monitoring workflows, not as a sole basis for clinical decisions."

{
"label": "Uncertain",
"raw_label": "Arrhythmia",
"confidence": 0.54,
"quality": { "ok": false,
"reasons": ["low_confidence"] }
}

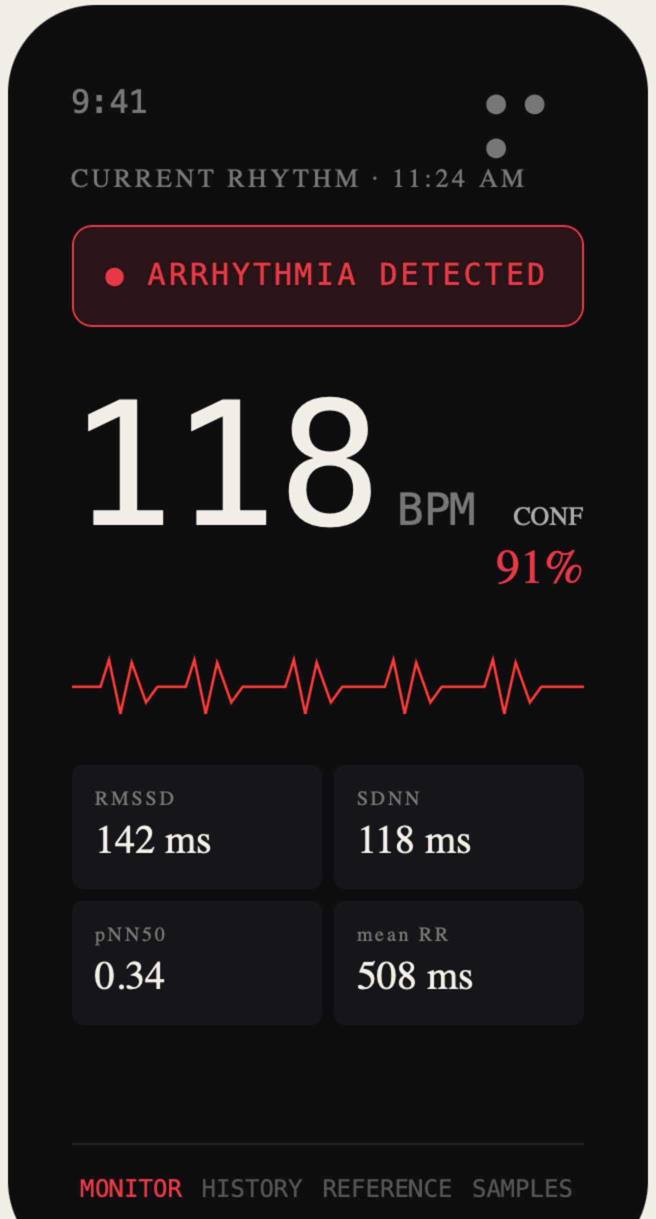
LAYER 04 / TWO VIEWS

One stream. Two interfaces.

Patient and clinician each get the view they need.

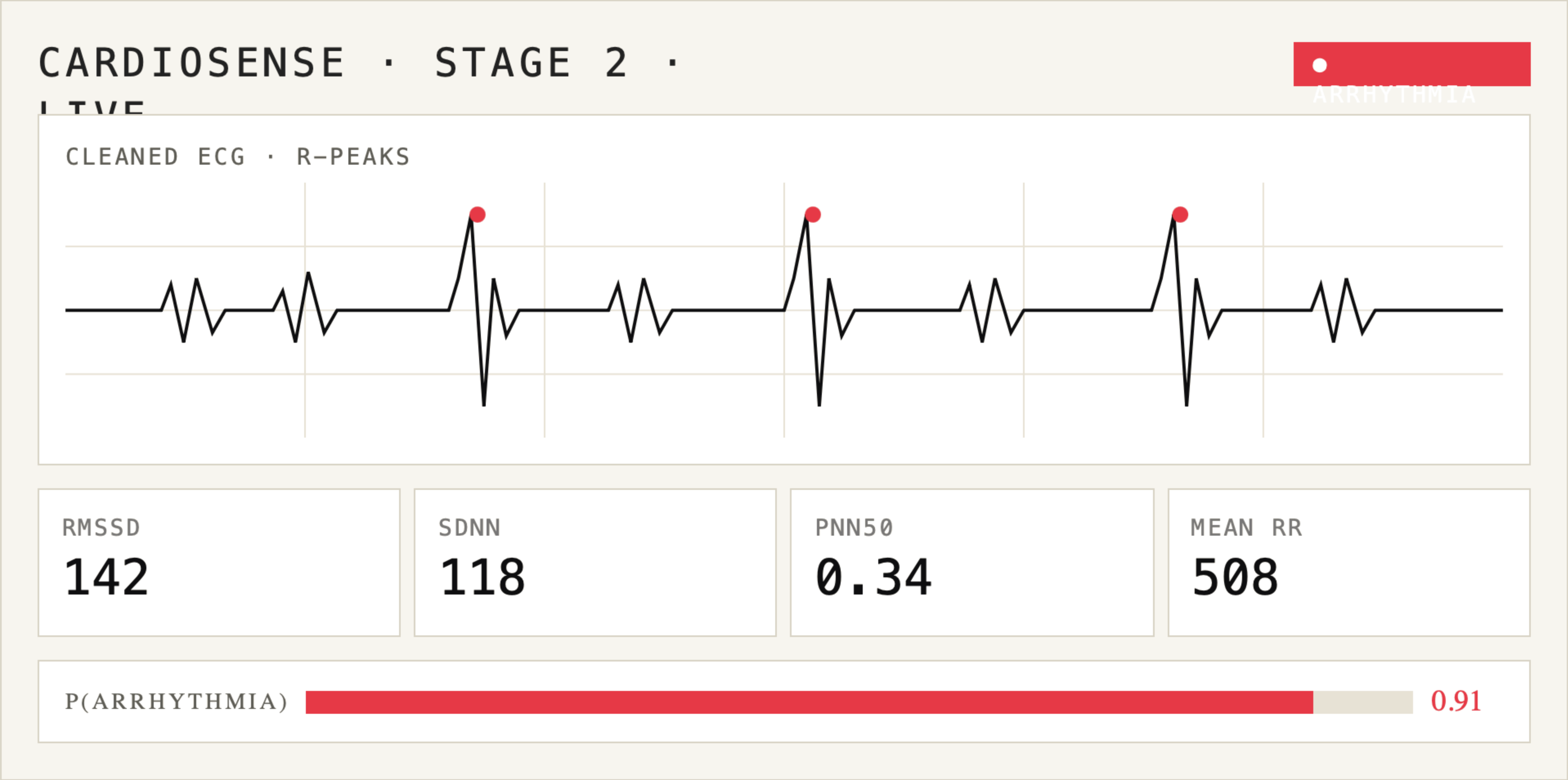
FOR THE PATIENT — IOS / SWIFTUI

Calm, plain-language readout. Polls every 2 s over local Wi-Fi.



FOR THE CLINIC — STREAMLIT DASHBOARD

Live waveform, R-peak markers, full HRV grid, class probabilities, and signal-quality flags.



$7.5B

global cardiac monitoring market.
Continuous monitoring is the fastest-growing segment, driven by aging populations and remote patient care.

THE GAP, IN ONE SENTENCE

No affordable, real-time, continuous arrhythmia detection exists outside a clinical setting.

EXISTING DEVICES

$300 – $1,500

ACCESS

physician's order

TURNAROUND

days to weeks

ADDRESSABLE

uninsured + rural

Hardware at cost. Subscription for continuity.

— 01

Hardware

Under $30 in materials at retail prices. Sold at or near cost. Lowers the entry barrier; community clinics can buy in volume without insurance reimbursement.

— 02

Software subscription

Continuous monitoring, clinical-grade event reporting, and a physician dashboard with longitudinal history. Tiered by clinic size.

— 03

Why margins improve at scale

The model is 200 KB. No cloud infrastructure. No per-inference compute cost. No PHI on the network. No data liability tail.

< $30

BILL OF MATERIALS

Compared to $300–$1,500 for existing patch monitors. At-scale procurement and an integrated AFE+MCU module would push this materially lower.

What CardioSense is not.

We are deliberate about what we are claiming.

→ NOT

FDA-cleared.

A research and monitoring tool. The iOS app states this explicitly in its Reference tab.

→ NOT

A replacement for a cardiologist.

The system surfaces patterns. Clinicians make decisions.

→ NOT

Trained on patch-derived data.

The model trained on MIT-BIH clinical recordings. Real-patch deployment requires additional validation.

→ NOT

A consumer fitness app.

No streaks. No gamification. No daily summaries. Medical software for serious use.

Built in 24 hours. Built for who needs it most.

Three students at UC Davis. End-to-end on May 9–10, 2026. Hardware, signal processing, training, inference, app, dashboards — none of it from an SDK or a paid API.

— HARDWARE & SIGNAL

EE lead

Analog front end design, electrode placement, BLE transport, raw-signal stream.

— MACHINE LEARNING

ML engineer

Dataset preparation, 1D-CNN architecture, training loop, record-level validation, uncertainty layer.

— PRODUCT & SOFTWARE

Software lead

SwiftUI app, signal-processing pipeline, Streamlit dashboards, demo orchestration, product design.

With acknowledgment to **Communicare Health Centers** (Yolo County) and **Clínica Tepati** (Sacramento) — community clinics that serve the populations this project is designed for, and the model for the kind of partner CardioSense is built to support.

— THANK YOU

A wearable that catches the arrhythmia your doctor missed — because it wasn't there when you walked in.

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github.com/arnavUCD/CardioSense

HACKDAVIS 2026

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